

How Breakable Is Privacy: Probing and Resisting Model Inversion Attacks in Collaborative Inference

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Abstract—Collaborative inference (CI) improves computational efficiency for edge devices by transmitting intermediate features to cloud models. However, this process inevitably exposes feature representations to model inversion attacks (MIAs), enabling unauthorized data reconstruction. Despite extensive research, there is no established criterion for assessing the difficulty of MIA implementation, leaving a fundamental question unanswered: *What factors truly and verifiably determine the attack's success in CI?* Moreover, existing defences lack the theoretical foundation described above, making it challenging to regulate feature information effectively while ensuring privacy and minimizing computational overhead. These shortcomings introduce three key challenges: theoretical gap, methodological limitation, and practical constraint.

To overcome these challenges, we propose the first theoretical criterion to assess MIA difficulty in CI, identifying mutual information, entropy, and effective information volume as key influencing factors. We use the Mutual Information Neural Estimator to quantify mutual information, a key metric for assessing the privacy-preserving effect of defences, and to validate the proposed criterion. Building on this insight, we propose SiftFunnel, a privacy-preserving framework to resist MIA while maintaining usability. Specifically, we incorporate linear and non-linear correlation constraints alongside label smoothing to suppress redundant information transmission, effectively balancing privacy and usability. To enhance deployability, the edge model adopts a funnel-shaped structure with attention mechanisms, strengthening privacy while reducing computational and storage burdens. Experiments show that, compared to state-of-the-art defence, SiftFunnel increases reconstruction error by $\sim 30\%$, lowers mutual and effective information metrics by $\geq 50\%$, and reduces edge burdens by almost $20\times$, while maintaining comparable usability.

Index Terms—Privacy-preserving collaborative inference, deep learning privacy, model inversion attack, information theory.

I. INTRODUCTION

THE rapid advancement of deep learning has led to increasingly complex neural networks that achieve remarkable performance in tasks like image recognition and

object detection. However, deploying such models on edge devices remains challenging due to limited computational resources [1]. Applications like autonomous driving, facial recognition, IoT surveillance, and remote healthcare require real-time processing, yet the limited computational capacity of edge devices makes standalone deep learning inference impractical [2]. Cloud computing provides substantial computational power but faces latency, bandwidth, and privacy concerns when transmitting raw data. Collaborative inference (CI) [3] mitigates these issues by partitioning neural networks into an edge model f_{edge} and a cloud model f_{cloud} , where f_{edge} extracts feature representations for f_{cloud} to perform inference. This method enhances efficiency while maintaining real-time performance and is widely applied in UAVs [4], IoT systems [2], and private cloud computing (PCC) [5]. With the rise of 5G, IoT, and AI, CI is poised for broader adoption.

However, recent studies have revealed significant privacy risks in CI, where input data on edge devices can be reconstructed by model inversion attacks (MIAs) [6], [7], [8], as shown in Figure 1. Unlike centralized deployments, where adversaries recover training data through model APIs [9], [10], CI often relies on shallow neural networks at the edge, producing feature representations that retain significant mutual information (MI) with input data [11]. This exposes raw inputs to adversaries, who can reconstruct sensitive data, such as facial images directly from transmitted features [12].

As shown in Figure 1, MIAs in CI can be categorized into maximum likelihood estimation-based (MLE-based) MIAs [6], [13] and generative model-based (Gen-based) MIAs [8], [13]. MLE-based attacks optimize input data to minimize the loss between the edge model's output and transmitted features but require white-box access, limiting practicality. Gen-based MIAs, on the other hand, use generative models to approximate the inverse mapping of the edge model, reconstructing inputs from features. Their success depends on data distribution alignment and the model's ability to learn the inverse function. These privacy risks have driven extensive research on attacks and defences, but existing methods still face notable limitations.

Firstly, existing research lacks a comprehensive analysis of the fundamental factors influencing MIA effectiveness in CI. Current attack methods focus on improving data reconstruction under specific assumptions, leveraging innovative techniques to enhance reconstruction fidelity and generalization. In recent years, defences have increasingly recognized the role of mutual information in MIAs and have shifted toward designing optimal information bottleneck (IB) approaches [14], [15] to

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Code is available at <https://github.com/SiftFunnel/SiftFunnel>.

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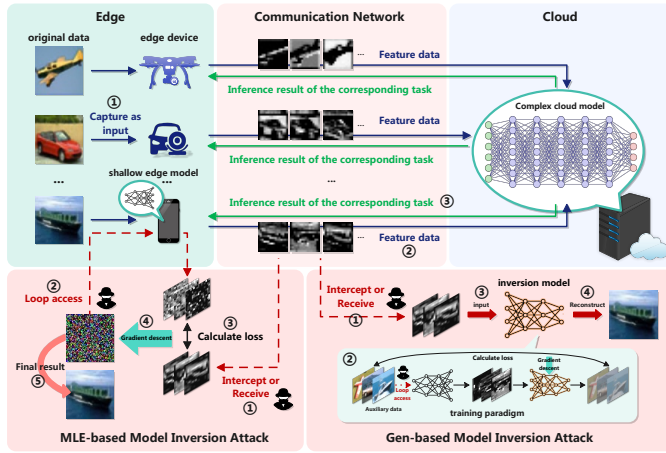


Fig. 1. Collaborative inference and the threat of model inversion attacks.

counter MIAs. However, these methods remain a methodological extension based on existing findings rather than advancing a fundamental theoretical understanding of MIAs. Consequently, their improvements are inherently constrained by existing technical limitations. Identifying the key factors that control MIA effectiveness and developing a systematic criterion for assessing attack difficulty would significantly deepen our theoretical insights and offer broader theoretical guidance for designing more effective defence strategies.

Secondly, the trade-off between usability, privacy, and deployability in CI is an underexplored and challenging area in current research. Encryption-based defences receive little attention in CI due to resource constraints, real-time demands [16], and vulnerability to side-channel attacks [17]. Mainstream MIA defence strategies in CI, constrained by a limited understanding of MIA effectiveness factors, primarily focus on reducing the mutual information $I(x; z)$ between inputs x and extracted features z to counter MIAs [14], [15], [11]. However, mutual information estimation is inherently challenging due to its reliance on accurate probability density estimation, large-scale training data, and the computational overhead associated with optimization techniques [14], [18], making the usability-privacy trade-off inefficient and difficult to guarantee. Moreover, some approaches compromise usability [13], increase computational and storage burdens on edge devices [14], and complicate further fine-tuning and optimization. Consequently, such methods are often less favoured in practical applications.

Thus, there is a pressing need for a novel approach grounded in an in-depth analysis of attack theories to provide interpretable and effective privacy protection while maintaining the usability and deployability of edge models. This need gives rise to three challenges. **Challenge 1 (Difficulty assessment):** How to establish a verifiable criterion for assessing MIA difficulty in CI? The fundamental factors determining MIA effectiveness remain poorly understood, making it difficult to develop a reliable assessment criterion. While MI's role in MIAs is recognized, its accurate estimation for validation remains challenging. **Challenge 2 (Methodological design):** How to effectively and interpretably remove redundant in-

formation from transmitted features while preserving edge model usability? Although IB theory provides a theoretically sound approach, its practical implementation is limited by the technical constraints of MI estimation. Current methods are computationally expensive and lack flexibility, making fine-tuning and further optimization difficult after initial deployment. **Challenge 3 (Practical constraint):** How to balance usability and privacy while ensuring practical deployability? Certain defence strategies [11] inevitably increase computational and storage burdens on edge devices. Furthermore, when applied to edge models designed for simple feature extraction, these methods struggle to counter advanced Gen-based MIAs effectively.

To overcome the above challenges, we proposed a criterion D_{mia} to evaluate MIA difficulty in CI and proposed a lightweight privacy-preserving strategy called Information Sifting Funnel (SiftFunnel), featuring 3 key aspects. **(1) Criterion basis:** SiftFunnel offers interpretable defence principles based on D_{mia} . We analyze MIAs via information theory, leveraging Fano's inequality and a PAC-privacy style bound to characterize when reconstruction becomes feasible under a fixed attacker model, linking inversion success to the residual uncertainty $H(x|z)$ and the information z carries about x . Motivated by this linkage, we propose an MIA difficulty criterion governed by four factors: $I(x; z)$, $H(x|z)$, $H(z)$, and the effective information mean $\delta(z)$ (the average number of non-zero elements in z). We validate these factors experimentally, reporting mutual information as a MINE-based estimate [19]. **(2) Usability-privacy trade-off:** SiftFunnel effectively reduces redundant information while preserving edge model usability. Aligning with D_{mia} to strengthen privacy protection, we aim to indirectly reduce $H(z)$ and $\delta(z)$ while increasing $H(x|z)$. Specifically, we incorporate distance correlation, Pearson correlation constraints, and a l_1 -norm constraint on z to enforce these objectives. This is more flexible and realizable than the direct estimation approach of previous work. To maintain usability, we integrate label smoothing and KL-divergence loss into the training process. This optimization enables the edge model to minify redundant information while ensuring the extraction of task-relevant features. **(3) Lightweight design:** SiftFunnel balances usability and privacy while light-weighting the edge model. We adopt a funnel-shaped edge model to reduce computational and storage overhead. Attention blocks enhance feature selection, while an upsampling module on the cloud side compensates for lost details, preserving inference accuracy. This design, guided by D_{mia} , also improves privacy by limiting feature $\delta(z)$ exposure.

Except for previous evaluation metrics, we incorporate estimated MI via MINE and $\delta(z)$ as metrics to assess defence effectiveness. Deployability is measured by the edge model's parameter count and inference latency. Systematic experiments confirm that SiftFunnel effectively balances usability, deployability, and privacy compared to existing methods, achieving superior MIA resistance while reducing mutual information and storage, with an average accuracy trade-off of 3%.

Our contributions are summarized as follows.

- We propose the first MIA difficulty criterion for CI. Leveraging Fano's inequality and a PAC-privacy style

bound, we relate reconstruction success to $H(x | z)$, $I(x; z)$ and $H(z)$, and identify $\delta(z)$ as an additional factor impacting attack effectiveness.

- We propose SiftFunnel, a privacy-preserving scheme for CI that constrains redundant features via linear and non-linear correlation, ensuring model usability with label smoothing and loss function regularization.
- We adopt a funnel-shaped structure with attention blocks in SiftFunnel to balance usability and privacy while keeping the edge model lightweight and deployment-friendly.
- We conduct systematic experiments to validate the proposed criterion and demonstrate SiftFunnel's effectiveness in balancing usability, privacy, and deployability. We additionally report estimated mutual information via MINE, $\delta(z)$, and deployment metrics such as parameter count and latency.

II. BACKGROUND AND RELATED WORK

This section provides background for our problem setting and contributions. We first summarize collaborative inference and its attack surface. We then describe representative model inversion attacks, including MLE-based and Gen-based methods. Finally, we review existing defences and their limitations in balancing privacy, usability, and deployability, which motivates our difficulty criterion and SiftFunnel.

A. Collaborative inference

Collaborative inference [3] has emerged as an effective solution. This approach divides the inference process into two stages: one part is performed locally on the edge device (which may consist of a single device or a chain of interconnected devices), while the remaining computation is uploaded to the cloud. Formally, this can be expressed as

$$f = f_c \circ f_e, \quad \text{i.e.,} \quad f(x) = f_c(f_e(x)). \quad (1)$$

where f is the fully trained model, x is the input, f_e represents the model deployed on the edge, and f_c corresponds to the portion deployed in the cloud.

CI typically begins with training f on centralized datasets in the cloud, then partitioning and deploying the f_e to the edge. Therefore, it is imperative to note that CI represents a scenario that has advanced to the inference application phase. Alternatively, approaches like Split Learning [20] allow edge devices to participate in model training locally. Regardless of the training approach, the inference process remains consistent. As depicted in Figure 1, the edge model computes an intermediate representation, which is transmitted to the cloud for completion of the inference process. The final output is either returned to the edge device or retained in the cloud for further analysis.

However, existing research [6] highlights that edge models are susceptible to MIAs, where adversaries reconstruct input data from transmitted features. Notably, in edge-based inference, computations can be distributed across multiple devices in a chained architecture [6], expressed as $f = f_{e1} \circ f_{e2} \dots \circ f_c$. Studies [13], [21] show that deeper neural networks closer to

the cloud transmit more complex, decision-focused features, making them more resistant to reconstruction. In contrast, the initial edge device generates features retaining more input information, making it highly susceptible to MIAs. This paper focuses on the MIA threat to the initial edge device, which employs a shallow neural network while the remaining computation occurs in the cloud. This common scenario in real-world applications presents a significant security risk due to the increased likelihood of input data reconstruction, highlighting the need for effective countermeasures.

B. Model inversion attack

Current MIAs in CI can be broadly categorized into MLE-based MIAs and Gen-based MIAs, with related research progress described as follows.

MLE-based MIAs. MLE-based MIA was first introduced by Fredrikson et al. [22], who reconstructed training data from a multilayer perceptron (MLP) by minimizing the difference between target and confidence scores using gradient descent. Subsequent research [6] incorporated Total Variation (TV) [23] loss as a regularization term to improve optimization, with these advanced techniques referred to as rMLE-based MIA. In this paper, both are collectively referred to as MLE-based MIA. In CI scenarios, MLE-based methods are particularly effective for reconstructing data from shallow neural networks, as the richer feature representations and lower complexity provide adversaries with detailed gradient information that is easier to optimize compared to the prediction vectors of deeper networks.

Gen-based MIAs. This approach was introduced by Yang et al. [24], who reconstructed input data in facial recognition systems by using generative deep learning models g to approximate the inverse mapping function of the target model f^{-1} , such that $g \approx f^{-1}$. By feeding prediction vectors into g , they successfully generated facial input images. Building on this approach, He and Yin et al. [6], [8] extended it to collaborative inference, demonstrating that input data from edge models is more vulnerable to reconstruction, thereby highlighting the risks. Unlike MLE-based methods, Gen-based MIA does not require white-box access to the edge model. Instead, it trains an attack model by calculating losses based on transmitted features and auxiliary data, enabling it to approximate the inverse mapping of non-linear computations. Furthermore, as edge models produce rich and redundant outputs, this facilitates the training of attack models and the reconstruction process, as analysed in detail in Section IV.

In recent years, driven by the rapid advances in generative models, new developments have emerged in generator-replacement attacks. Specifically, some works have leveraged generative adversarial networks (GANs) [25], [26], [27], while others have explored diffusion-based models [28], [29] to improve reconstruction fidelity. Despite architectural innovations in the employed generators, the fundamental principle remains unchanged, i.e., approximating the inverse mapping of the target model. Nevertheless, the training of such generators generally presupposes white-box access, rendering them inflexible and not adequately accounting for adaptive attacks.

Furthermore, their effectiveness is still subject to the same inherent limitations, i.e., the reliance on intermediate-layer transmitted features as inputs or conditions, which does not alter the theoretical analysis presented in this paper.

Some studies [25], [27] have referred to such attacks as *Data reconstruction attacks*. Nevertheless, their underlying principle remains consistent with MIA [24] and the subsequent work of He et al. [6]. For the sake of conceptual consistency with the majority of prior research, this paper continues to adopt the term *Model Inversion Attack*.

Facts and Motivation. Although MIA may be confused with *Membership Inference Attack* [30], its meaning is clearly defined in this work and widely adopted in the model inversion literature. Moreover, the feasibility of MIA and their threat to CI have been well established. While encryption may protect data in transit, it does not prevent input reconstruction once intermediate features are exposed. Such exposure can occur locally or via compromised cloud infrastructure. Moreover, encryption adds substantial overhead to edge systems, limiting its practicality. These concerns highlight the need to analyse attack factors and design lightweight, deployable defences.

C. Related defence work

To safeguard input data collected by edge devices in CI from potential leakage, He et al. [6], [13] emphasized the importance of developing effective defence strategies while identifying these vulnerabilities. Existing defences can be classified into three main categories: perturbation-based, IB-based, and neural network depth (NND) & IB-based approaches. The following sections outline these strategies.

Perturbation-based defence. This approach introduces carefully designed noise perturbations into the gradients for training or trained edge model output features to reduce the likelihood of adversaries extracting meaningful information and defending against MIA. He et al. [13] initially proposed using Gaussian noise and random feature dropout to limit the effective information accessible to attackers, and later, Wang et al. [31] extended this strategy by dynamically adjusting the depth of the edge model while adding differential privacy noise to transmitted features. Although these methods enhance defences, they significantly impact model usability, typically reducing accuracy by approximately 10%, and remain less effective against Gen-based MIA. The root cause is that the decoder in Gen-based MIAs is sufficiently expressive to adapt to noise or feature dropout, similar to the training of a denoising auto-encoder (DAE) [32], potentially increasing attack robustness. In centralized MIA defences, Struppek et al. [33] proposed perturbing training labels with a negative label smoothing factor, significantly reducing MIA effectiveness and offering a novel perspective for designing defence strategies.

IB-based defence. This approach enhances defences against inference attacks by limiting the mutual information between input data x and intermediate features z while preserving the mutual information between z and outputs y , thus maintaining inference performance [34]. The objective of this defence can be formally expressed as $\min_{\theta} -I(z; y) + I(x; z)$. Initially,

Wang et al. [34] introduced information bottleneck theory into MIA defences, but their method was not optimized for CI and required substantial computational resources for intermediate feature processing. To address this, Peng et al. [15] proposed BiDO (Bilateral Dependency Optimization), which uses dependency algorithms to optimize correlations among inputs, features, and outputs. BiDO reduces input-feature correlation while enhancing feature-output correlation, achieving a balance between defence and performance. Additionally, Wang et al. [14] and Duan et al. [18] explored mutual information estimation between x and z using methods based on mutual information definition and the CLUB (Contrastive Log Upper Bound) [35] technique, incorporating the estimates as constraints in the training loss function. IB-based defences provide interpretable protection mechanisms and valuable design insights for future research. However, these methods face limitations: BiDO relies on kernel functions for non-linear correlation constraints, with performance highly sensitive to hyper-parameters like kernel bandwidth; poor parameter selection can lead to suboptimal results. Mutual information estimation based defences require explicitly estimating the joint distribution and marginals from finite mini-batch samples, which is difficult in high dimensional continuous spaces and therefore suffers from large estimation error unless the sample size and batch size are sufficiently large.

NND & IB-based defence. Leveraging deeper networks and IB theory, this approach enhances defence effectiveness with targeted loss functions. Ding et al. [12] proposed increasing network depth while compressing newly added hidden layers to restrict redundant information transmission. Subsequently, Azizian et al. [11] further improved defences by introducing an auto-encoder (AE) structure, with encoders at the edge and decoders in the cloud. The edge encoder increases network depth and reduces feature dimensionality to achieve the IB effect, while the cloud decoder restores information to maintain predictive performance while applying a l_1 -norm constraint to reduce attack risks. Adversarial training [12] is also incorporated to strengthen robustness against MIAs.

However, this method has certain limitations. Increasing network depth and adding modules raise computational and hardware demands on edge devices, limiting applicability. Adversarial training, while strengthening defences, introduces additional challenges, it is less effective against advanced attacks, complicates fine-tuning, and degrades inference performance [36]. Additionally, adversaries can adapt their attack models using similar adversarial strategies, weakening defence effectiveness [37]. While these methods improve defences, their implementation requires careful consideration of computational resources, model adaptability, and application suitability. Moreover, an open question remains: does the primary defensive effect arise from increased architectural depth itself, or is it mainly attributable to adversarial training? SiftFunnel enhances computational efficiency by deepening edge model structures and adopting a lightweight design, exhibiting IB-like characteristics after training. Thus, our method falls under the NND & IB-based defence, and its effectiveness is empirically validated through ablation studies on both the proposed correlation constraints and the model design.

III. THREAT MODEL

This section defines three attack scenarios, considering the implications of adaptive attacks. It outlines the adversary's shared knowledge, specifies and explains the model and data types on which this work focuses, and details the attack strategies employed under each scenario.

A. Knowledge background of adversary

Attack scenarios. Based on the adversary's level of knowledge, attack scenarios can be categorized into the following three types:

- 1) **White-box Scenario:** Assume the adversary is the service provider or a cloud-platform administrator who can access model files before and after deployment. In this scenario, the adversary has complete white-box knowledge of the edge model, including its architecture, parameters, and gradients, and is able to back up and export the full set of parameters and structure deployed at the edge.
- 2) **Black-box Scenario:** Assume the adversary is an ordinary end user or an application caller with only black-box access to the model. This restricts their capability to observe the intermediate features, and their understanding of the edge model's architecture is limited to prior knowledge or incomplete assumptions.
- 3) **Gray-box Scenario:** Assume the adversary is a partner developer or third-party maintainer who knows parts of the model architecture, for example from published papers or earlier versions, but does not know defensive modifications applied during edge deployment such as added hidden layers.

The gray-box scenario enables the adversary to leverage their partial understanding of the edge model's architecture to enhance the capability of their attack model. For instance, if the defender partitions the first three convolutional layers of a ResNet-18 model [38] and further deepens the network, while the adversary is only aware of the original ResNet-18 architecture, they can only construct an approximate attack model based on partial knowledge of the initial three layers. However, adversaries can address this limitation in this scenario by utilizing their understanding of the edge model's architecture and reasoning about its mapping capabilities to refine their attack model. This setting is also our consideration of adaptive attacks in this work.

Knowledge shared by adversaries in different scenarios.

The adversary is assumed to understand the edge model's task and enhances their attack by training Gen-based MIA on a closely matching data distribution (Section VI-A). This work focuses on the model's inherent privacy issues through intermediate features. Accordingly, the adversary is assumed to observe the transmitted intermediate features and their dimensional distribution without transmission noise or additional perturbations, which facilitates optimizing the MIA loss function and training the attack model. This setting reflects the attacker's best-case effectiveness of MIA and ensures that the measured privacy protection is attributed to the proposed

defence and model design, rather than being assisted by transmission noise.

This study focuses on image classification but emphasizes that the defence methods apply to other tasks where the edge model primarily handles feature extraction. Since segmentation, detection, and classification also rely on edge-processed features, MIA and defences are task-agnostic. Prior work [11] has demonstrated input reconstruction and defence effectiveness in object detection using Gen-based MIA.

Moreover, image data is particularly vulnerable to MIAs due to its high dimensionality, as shallow convolutional layers retain more redundant information [39], [38]. This increases the challenge of protecting image data, making it an ideal test case for privacy defences. In contrast, textual data, with its strong contextual dependencies and syntactic complexity, has shown limited success in achieving high-quality MIA reconstructions [40], [41]. This study focuses on image data due to its interpretability, making it suitable for evaluating privacy defences.

B. Attack strategy of the adversary

In the white-box attack scenario, the adversary can perform attacks using the MLE-based MIA method proposed in [6], [13]. To ensure the experiments are both comparative and representative, hyper-parameter adjustments were made to achieve performance closer to optimal.

In the black-box attack scenario, the adversary can employ methods from [24], [6], [8], relying solely on output features for reverse mapping without access to the edge model's internal details. When protected by NND & IB-based defences, the adversary lacks knowledge of architectural changes and must design the attack model based on prior assumptions.

In the gray-box attack scenario, the adversary can employ black-box strategies while leveraging known architectural information to optimize the attack model. This allows the reverse mapping capability to adapt to changes in the edge model. Detailed designs and training procedures for the attack models are provided in Section VI.

Although recent GAN and diffusion-based attacks [25], [28] offer valuable perspectives, they are not included in our experimental evaluation. This decision is made because their training still relies on white-box settings, suffers from limited flexibility, and faces similar implementation constraints as prior approaches. Moreover, the primary focus of this paper is to compare the effectiveness of different defence mechanisms.

IV. THEORETICAL ANALYSIS OF MIA IN CI

This section analyses key factors affecting MIA effectiveness in CI, introduces a formulaic representation of its principles, and establishes a difficulty criterion from an information-theoretic perspective. We validate this criterion through experiments and use it to guide the design of SiftFunnel.

A. Principles analysis of MIA

MIAs are categorized into MLE-based and Gen-based methods, both aiming to reconstruct input data using the output of

the edge model. MLE-based MIA calculates the loss between transmitted features and the output, then iteratively applies gradient descent to optimize the input, achieving reconstruction of the original data. The attack principle is formulated as follows:

$$\begin{aligned} x^{(k+1)} &= x^{(k)} - \eta \nabla_x \mathcal{L}(x, f_{\text{edge}}(x^{(k)}), z), \\ x^* &= \arg \min_x \mathcal{L}(x, f_{\text{edge}}(x^{(k)}), z) \end{aligned} \quad (2)$$

Where $x^{(k)}$ represents the input value at the k -th iteration, typically initialized as Gaussian noise, zeros, or ones. x^* denotes the final reconstructed result, z represents the transmitted features of the target input. $f_{\text{edge}}(\cdot)$ refers to the target edge model, $\nabla_x \mathcal{L}$ is the gradient calculated from the loss function and edge model parameters to optimize x , and η is the step size for each update.

According to [6], the loss function combines the l_2 distance between $f_{\text{edge}}(x^{(k)})$ and z with a TV regularization term for x , forming the basis of rMLE-based MIA. The method relies on gradient descent, where the loss values depend on output features, and gradients are computed using the target model's parameters.

To achieve effective attacks, sufficient output feature information $\delta(z)$ and a well-chosen loss function are essential to ensure optimization space. We quantify the effective output information by the number of non-zero feature elements in z , defined as

$$\delta(z) \triangleq \mathbb{E}_X \left[\sum_{j=1}^{d_z} \mathbf{1}(z_j \neq 0) \right], \quad (3)$$

where d_z is the total number of elements in z after flattening.

Additionally, white-box knowledge of the edge model is required, as complex model mappings and limited feature information increase the risk of gradient descent converging to local minima [10], [39]. For instance, if two similar images produce nearly indistinguishable output features, reconstruction becomes challenging.

When dealing with complex edge model mappings or models protected by defence techniques, directly computing gradients in MLE-based MIA can significantly hinder optimization performance. To address this, He et al. [6], [13], building on the method from [24], developed a reverse mapping network for the edge model, which is categorized as Gen-based MIA. The key idea is to use a generator to approximate the reverse mapping of the edge model. Different methods adopt various strategies to train the generator. For instance, Yang et al. [24] trained a decoder using auxiliary task-related datasets, while Yin et al. [8] utilized intercepted feature data for training. The principle of this attack is summarized as follows.

$$\begin{aligned} G &= \arg \min_G \mathbb{E}_{x' \sim p(x')} \mathcal{L}(G(f_{\text{edge}}(x')), x'), \\ x^* &= G(z) \end{aligned} \quad (4)$$

Where x' represents samples from an auxiliary data distribution $p(x')$. The loss function, typically mean square error, measures the distance between x' and the reconstructed data $G(f_{\text{edge}}(x'))$ or $z = f_{\text{edge}}(x')$. The trained generator G must achieve the desired mapping $G \approx f_{\text{edge}}^{-1}$ to reconstruct x^* .

The above formula highlights the conditions required for effective attacks:

- 1) A well-matched training dataset aligned with the target task, ideally with a distribution similar to the target dataset [24].
- 2) Sufficient and highly separable input information for the generator.

The rationale is that if the information in $f_{\text{edge}}(x')$ is insufficient or cannot independently represent x' 's features, the loss in (4) will struggle to converge as $G(f_{\text{edge}}(x'))$ remains ambiguous, limiting its training. Among them, the maximum separability is more decisive. For instance, even with random feature drops, if the remaining features distinguish inputs well, the attack can still reconstruct effectively [13]. Such frameworks resemble DAE [32], where the decoder can adapt to noise or feature drops.

Gen-based MIA demonstrates superior reconstruction performance compared to MLE-based MIA in defence scenarios. First, it avoids relying on model parameters for gradient computation, eliminating challenges from non-convex optimization. Second, it depends only on the output of the edge model, offering greater flexibility. Furthermore, existing defences struggle to effectively remove redundant feature information, allowing Gen-based MIA to overcome defences if the generator is capable of learning the edge model's reverse mapping.

B. Difficulty criterion for MIA in CI

We summarized the factors influencing MIA effectiveness using a formulaic representation. To further evaluate the difficulty of MIA implementation, we will formalize and analyse these factors from an information-theoretic perspective to establish a clear criterion.

First, the mutual information between the input x and the feature z , $I(x; z)$, can be expressed as:

$$I(x; z) = H(x) - H(x | z) = H(z) - H(z | x) = H(z). \quad (5)$$

Where $H(x|z)$ denotes the conditional entropy of the discrete input x given the feature z , and $H(z|x)$ denotes the reverse. Throughout, we interpret $H(\cdot)$ as the discrete Shannon entropy because both x and the transmitted feature z are represented under finite precision at run time (e.g., pixel quantization and FP rounding / INT quantization in communication and storage). Under deterministic inference on the edge (i.e., f_{edge} with deterministic finite-precision arithmetic), the mapping from x to z has no intrinsic uncertainty, so $H(z|x) = 0$ and hence $I(x; z) = H(z)$.

This assumption holds primarily because the CI scenario involves deterministic inference, where model parameters are fixed and the feature extractor contains no stochastic components. Even if x is perturbed by noise, the mapping $f_{\text{edge}}(x)$ still produces a unique and deterministic output z , ensuring that $H(z|x) = 0$. Moreover, as stated in Section III-A, this analysis focuses on inherent privacy issues, so we define z as the unperturbed intermediate feature and do not model transmission noise, so that this premise remains valid.

Factors influencing MLE-based MIA. From Equation (2) and (5), we observe that complex mappings and information loss during neural network forward propagation reduce $I(x; z)$ and increase $H(x|z)$, making non-convex optimization with backpropagation more challenging. Consequently, reconstructing x from z becomes more difficult for MLE-based MIA.

These methods primarily compare z with generated features, but a poorly trained or protected model can produce z with high $\delta(z)$ yet low $H(z)$, indicating suboptimal separability. This weakens attack effectiveness despite the apparent information sufficiency. Thus, MLE-based MIA difficulty is more influenced by $I(x; z)$, $H(x|z)$, and $H(z)$.

To theoretically assess the difficulty of recovering the input x , we consider the reconstruction error probability P_e . In information theory, Fano's inequality provides a fundamental lower bound on the reconstruction error in terms of conditional entropy. Modelling the MIA process as a Markov chain $x \rightarrow z \rightarrow x^*$, where x^* is the reconstruction from z , Fano's inequality [42], [43] implies:

$$H(P_e) + P_e \log |\mathbb{X}| \geq H(x | x^*) \geq H(x | z) \quad (6)$$

where $\mathbb{X}$ denotes the value space of x .

Since $|\mathbb{X}| \geq 2$ and $H(P_e) \leq 1$, we can derive a simplified lower bound:

$$\begin{aligned} P_e &\geq \frac{H(x | z) - 1}{\log |\mathbb{X}|} = \frac{H(x) - I(x; z) - 1}{\log |\mathbb{X}|} \\ &= \frac{H(x) - H(z) - 1}{\log |\mathbb{X}|} \end{aligned} \quad (7)$$

As $H(x)$ and $|\mathbb{X}|$ are intrinsic to the dataset and task, they typically remain fixed. Therefore, increasing the lower bound of P_e , and hence the difficulty of model inversion, is primarily influenced by $H(x | z)$, $H(z)$, and $I(x; z)$.

Although Fano's inequality is often presented for exact recovery, it only requires a binary success criterion and therefore remains applicable under relaxed semantic definitions of success. We can declare an inversion successful iff $\text{MSE}(x, x^*) \leq \tau$, and define

$$P_e(\tau) \triangleq \Pr[\text{MSE}(X, X^*) > \tau].$$

We set $\tau = 0.02$ based on empirical calibration in Section VI-C. With this definition, Fano's inequality still holds and yields a global lower bound on $P_e(\tau)$ that is monotone in mutual information and entropy.

While Fano provides a global floor, it can be loose under non-uniform priors. We therefore integrate probably approximately correct (PAC) privacy guarantees [44], [45], [46], which yield a prior-aware interval for the posterior error after observing Z , rather than only a one-sided lower bound. According to Corollary 1 in [44], the posterior error after observing Z is bounded relative to the prior error as follows:

$$\begin{aligned} P_e^{\text{post}}(\tau) &\geq \max\left\{0, P_e^{\text{prior}}(\tau) - \sqrt{\frac{I(X; Z)}{2}}\right\}, \\ P_e^{\text{post}}(\tau) &\leq \min\left\{1, P_e^{\text{prior}}(\tau) + \sqrt{\frac{I(X; Z)}{2}}\right\}. \end{aligned} \quad (8)$$

Here $P_e^{\text{prior}}(\tau)$ is determined by the adversary's prior knowledge (e.g., target model architecture and training data); it also

anchors the PAC bound, acting as the centre of the interval of radius $\sqrt{\frac{I(X; Z)}{2}}$ and thereby determining the attainable range of $P_e^{\text{post}}(\tau)$ for a given $I(X; Z)$. In particular, more informative priors decrease $P_e^{\text{prior}}(\tau)$ and permit a lower $P_e^{\text{post}}(\tau)$ under the bound, whereas restricting such information increases $P_e^{\text{prior}}(\tau)$ and limits the best-case reduction.

Together, the two bounds serve different purposes. The Fano bound provides a prior-agnostic monotone relationship that reveals how information-theoretic factors affect inversion difficulty. The PAC interval further shows that even an optimally trained attacker is constrained by both the information content $I(X; Z)$ and the attacker's prior knowledge: it upper- and lower-bounds the attainable posterior error, so inversion performance cannot improve arbitrarily beyond what the $(I(X; Z), P_e^{\text{prior}}(\tau))$ pair permits.

Factors influencing Gen-based MIA. From Equation (4) and (5), Gen-based MIA requires sufficient input information $\delta(z)$ and high $H(z)$ to ensure maximum separability and enable loss convergence, allowing the generator to simulate reverse mapping effectively. A larger $\delta(z)$ not only facilitates generator training but also directly impacts the reconstruction of x^* . At the same time, the relationship in Inequality (7) and (8) also holds here [47]. Therefore, in addition to the factors influencing MLE-based MIA, Gen-based MIA is further effectively affected by $\delta(z)$.

Criterion formula. The analysis indicates that the difficulty of implementing both types of MIA is inversely proportional to the mutual information, entropy, and output information. Therefore, the difficulty criterion for MIA can be formulated as follows, where k_1 and k_2 are proportionality constants:

$$D_{\text{mia}} \propto \frac{H(x | z)}{H(z)^{k_1} \cdot \delta(z)^{k_2}}, \text{ with } I(x; z) = H(z) \quad (9)$$

The above Fano- and PAC-based analyses make the title notion of privacy *breakability* operational: they characterize when inversion from z to x becomes feasible under a fixed success criterion and attacker model. Eq. (9) further compresses the dominant factors into a composite difficulty proxy; as such, it provides an inverse reflection of breakability.

Based on this criterion, existing defences [34], [15], [18] using the information bottleneck theory can be understood as approximating a reduction in $I(x; z)$ through various algorithms to increase the difficulty of MIA. Perturbation-based defences [13] aim to directly increase $H(x | z)$. The defensive effect of deeper neural networks [12], [11] can be explained by their ability to reduce $I(x; z)$ and $\delta(z)$. This is mainly because, without skip connections [48], mutual information progressively decreases during forward propagation in neural networks. In contrast, residual networks with skip connections preserve or even enhance $\delta(z)$ by adding layer outputs to inputs, but the decline in mutual information still poses challenges for MIA (see Section IV-C for experimental cases).

C. Demonstration and Guidance for defence

To validate D_{mia} and the theoretical analysis, we designed experiments using a CNN that progressively reduces the spatial size while increasing the channel size, based on the framework

in [24], [38], and ResNet-18 with skip connections, quantifying each block. We trained MINE [19] on the same dataset used to train the full model f to estimate the lower bound of $I(x; z)$. The core idea of MINE is to optimize a neural network estimator using the Donsker-Varadhan (DV) representation [49] as the loss function, providing an approximate estimate of the mutual information between random variables.

Mutual information is defined as follows:

$$I(X; Z) = \mathbb{E}_{p(x,z)} \left[\log \frac{p(x, z)}{p(x)p(z)} \right]. \quad (10)$$

Here $p(x, z)$ is the unknown joint distribution and $p(x)$ and $p(z)$ are the unknown marginals. Therefore, computing $I(X; Z)$ from the definition requires estimating these distributions, or equivalently the density ratio $p(x, z)/(p(x)p(z))$, which is statistically and computationally intractable for high dimensional representations. In practice, existing defence methods can only approximate the expectations using finite-sample, mini-batch Monte Carlo [50] estimates or simplified assumptions, and limited batch sizes often introduce non-negligible variance and instability in the resulting estimates.

To avoid explicit distribution estimation, the DV representation reformulates mutual information as a variational optimization problem:

$$I(X; Z) \geq \sup_T \mathbb{E}_{p(x,z)} [T(x, z)] - \log \mathbb{E}_{p(x)p(z)} [e^{T(x,z)}], \quad (11)$$

Where T is a measurable function. This formulation enables estimating $I(X; Z)$ from samples by optimizing T to maximize the lower bound, using mini-batch approximations of the expectations over $p(x, z)$ and $p(x)p(z)$.

MINE achieves this by parameterizing T with a neural network T_θ and maximizing the DV bound through gradient-based optimization:

$$\hat{I}(X; Z) = \mathbb{E}_{p(x,z)} [T_\theta(x, z)] - \log \mathbb{E}_{p(x)p(z)} [e^{T_\theta(x,z)}]. \quad (12)$$

Where $\hat{I}(X; Z)$ is the estimated DV lower bound. Since the second term involves an exponential moment, the estimator can be sensitive to mini-batch noise; using sufficiently large batches improves the stability. In our experiments, we fix the MINE batch size to 128.

As shown in Figure 2, we trained the target models on the CIFAR-10 [51] and used Equation (12) to train MINE for the corresponding blocks. Figures 2 (a, b) show that increasing network depth reduces both mutual information and effective information, leading to a decrease in attack performance. Figures 2 (c, d) illustrate that while skip connections within two layers do not reduce $\delta(z)$, they slightly lower $I(x; z)$, moderately affecting attack performance. Previous research [48] identified vulnerabilities caused by skip connections in centralized MIA. Our study further highlights their impact on MIA effectiveness in CI.

However, transmission between blocks in ResNet alters feature channels and dimensions, requiring a shortcut layer, leading to a more pronounced decline in mutual information compared to direct stacking within layers. This, in turn, degrades MIA performance and aligns with our criterion.

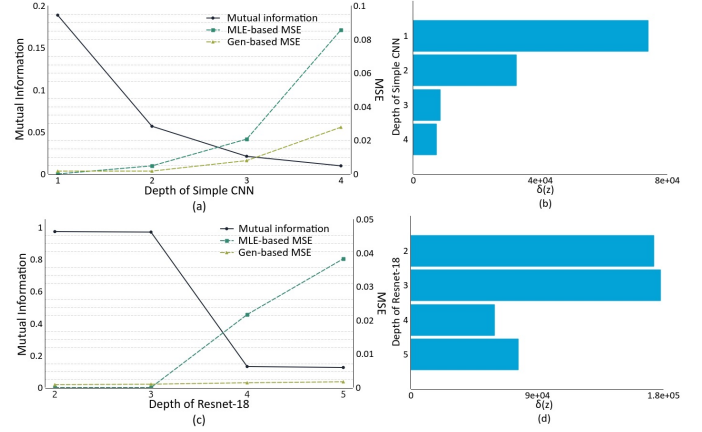


Fig. 2. The impact of neural network depth on $I(x; z)$, $\delta(z)$, as well as MLE & Gen-based MIA.

Notably, while mutual information at ResNet's fourth layer is higher than that at CNN's second layer, MLE-based MIA does not perform better. This stems from the fact that these methods rely heavily on gradient descent, making them sensitive to suboptimal optimizers, hyper-parameter settings, and deeper network structures, which hinder gradient calculations for optimizing x . The lack of flexible and efficient methods for MLE-based MIA remains a gap in current research, often leading to their exclusion in defence evaluations—a topic warranting further exploration [12]. In our experiment and Section VI, we still evaluate MLE-based MIA against various defences and provide evidence demonstrating the effectiveness of our method in defending against such attacks.

In conclusion, through theoretical analysis and experimental validation, we proposed D_{mia} and determined that effective defence strategies must impact mutual information, entropy, and $\delta(z)$. The following considerations can guide defence design:

- Since $I(x; z)$ is challenging to compute and current algorithms lack flexibility and efficiency, defences should focus on indirectly influencing $H(x | z)$ and $H(z)$.
- The design of edge models should strategically affect $H(z)$ and $\delta(z)$.
- For practical applications, the light-weighting of edge models should balance usability with their impact on the factors outlined in the criterion.

V. SIFTFUNNEL: INFORMATION SIFTING FUNNEL

Building on the theoretical analysis, criterion, and experiments in Section IV, this section introduces SiftFunnel and its adherence to D_{mia} in edge model and loss function design. We first outline the core principles and objectives of SiftFunnel. Then, we detail the edge model's structure, key modules, and role in the forward propagation. Finally, we explain the loss function's impact on $H(x|z)$, $H(z)$ and $\delta(z)$, highlighting its ability to extract useful information while minimizing redundancy and increasing the uncertainty of z to x .

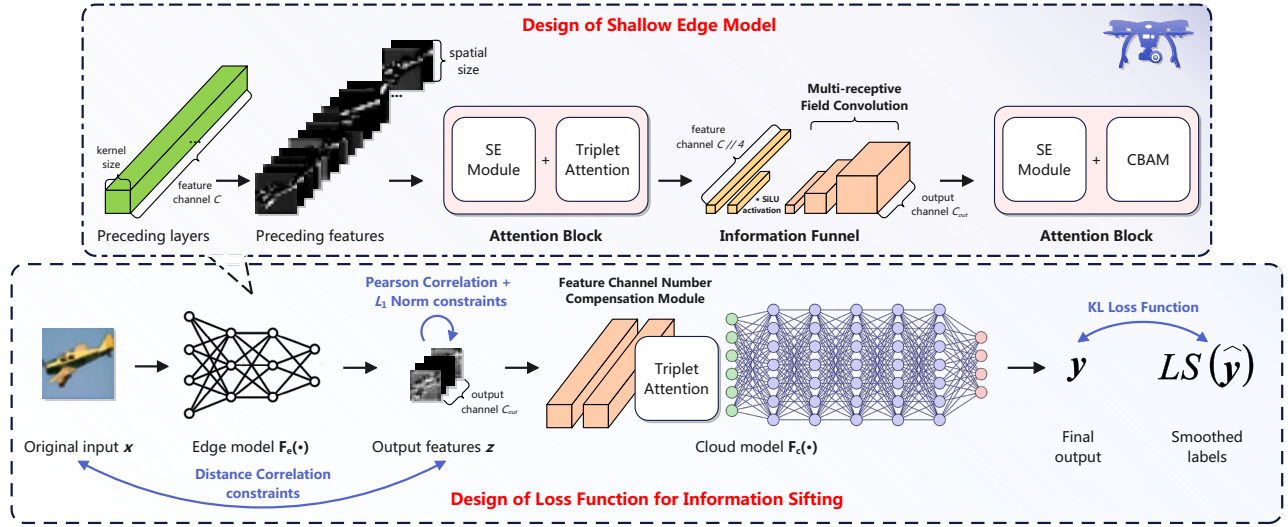


Fig. 3. Overview of SiftFunnel. Specifically, the design is divided into two aspects: the architecture and the loss function for the edge and cloud model.

A. Overview of SiftFunnel

Guided by D_{mia} , SiftFunnel balances usability and privacy by increasing uncertainty in inferring x from z while limiting redundant information and separability. As shown in Figure 3, SiftFunnel introduces two key improvements: First, the edge model modifies the final hidden layer into a funnel-shaped structure and incorporates attention modules. Second, the loss function is optimized to ensure that model training balances usability with privacy protection.

Specifically, in the edge model design, the final layer adopts a funnel-shaped structure, reducing feature channels while preserving spatial dimensions. This reduces computation, lightens the edge device load, and limits redundant information. Attention mechanisms and multi-receptive field convolutions enhance task-relevant information, while a cloud-based up-sampling module preserves accuracy. To reduce redundancy and $I(x; z)$, and increase the uncertainty of z to x , we use Distance Correlation [52] for nonlinear correlations and the Pearson Coefficient [53] with l_1 -norm for linear correlations. Additionally, Label Smoothing (LS) and Kullback-Leibler (KL) divergence [49] are used during training to improve model usability while reducing the separability of z .

B. The Design of Shallow Edge Model

Information Funnel. Deepening the neural network enhances feature extraction and reduces $I(x; z)$, as confirmed in Section IV. Previous work [21] shows through Fisher information [54] that deeper networks improve task relevance and resist MIA. Inspired by this, we redesign the final layer of the edge model with an Information Funnel module. This module reduces output channels to one-fourth in the first layer and further in the second layer, with channel selection based on the spatial size of feature maps. If the spatial size is reduced significantly, channels can exceed 10 to maintain accuracy. If unchanged, channels can be set to 2 to limit redundancy while preserving accuracy. Based on Section IV's analysis, the channel count has a predictable impact on utility and

privacy: per Eq. (9), a higher channel width directly increases $\delta(z)$ and indirectly decreases $H(x|z)$, improving utility at the cost of privacy, whereas a lower channel width does the opposite. Therefore, a cloud deployment can adaptively set this value according to the desired privacy level. In Section VI-C (item 4), our no-funnel ablation gives an empirical lower bound on architecture-level privacy protection, and we also report channel-count ablations for both spatial-size-reduced and spatial-size-unchanged settings.

Adjusting channels based on spatial size is necessary because spatial information loss is irreversible, whereas channel information loss can be compensated in the cloud model. This compensation increases $I(z; y)$ without reversing the reduction in $I(x; z)$, maintaining high attack difficulty as shown in Section VI-C. So, we add the channel compensation module in the cloud model to guarantee usability. Importantly, assumptions about extracting features from non-independent transmission modules are idealized, requiring adversaries to have full white-box knowledge of the cloud model as if they were the developer; we also explicitly evaluate this adaptive attack in Table VIII.

The Information Funnel's channel structure controls output but does not limit redundancy or enhance task-relevant information transmission, affecting usability. Inspired by Inception networks [55], we redesign it as a multi-receptive field convolutional framework with three low-channel layers. A 1×1 kernel fuses channel information, followed by 3×3 and 5×5 kernels to expand the receptive field, extract richer task-relevant features, improve decision-making, and reduce mutual information with x , while lowering parameters of the edge model and controlling $\delta(z)$.

Attention Block. To enhance the output of task-relevant information from the Information Funnel, attention modules are incorporated both before and after it. The first block combines an SE block [56] and Triplet Attention [57]. The SE block evaluates the contribution of information within each channel, suppressing channels with low task relevance. Triplet Attention reduces spatial information while main-

taining smooth, continuous outputs, preserving fine-grained features for further processing. The second block integrates SE and CBAM modules [58]. Compared to Triplet Attention, CBAM is a lightweight module that sequentially calculates channel and spatial weights to enhance or suppress input features. Since CBAM's channel and spatial attention are independently designed, certain weights are suppressed redundantly, resulting in sparser outputs. This sparsity is beneficial for output processing, effectively increasing $H(x | z)$ while reducing $H(z)$ and $\delta(z)$.

Similarly, the impact of attention mechanism is discussed in Section VI-C, where our no-attention ablation provides an empirical lower bound on utility and privacy protection.

C. The Design of Loss Function

To further enhance model usability and data privacy, appropriately designed loss constraints must be applied to x and z to reasonably increase $H(x | z)$ and suppress $H(z)$ within the edge model structure. Without intervention, neural networks may retain irrelevant input information in their output during forward propagation. This information, which does not significantly impact decision-making, reduces the difficulty of MIAs by maintaining the mutual information and entropy. The key challenge is how to achieve this effectively and flexibly.

Estimating $I(X; Z)$ for high dimensional representations is statistically challenging and can be unstable under finite-sample, mini-batch approximations, as it depends on unknown joint and marginal distributions. As a result, prior defences use more tractable surrogates for dependence as proxies for MI, such as covariance-based correlation measures.

Some defence approaches utilize constrained covariance (COCO) [59] to measure dependence, which primarily captures linear relationships between x and z . However, since neural networks predominantly involve nonlinear transformations, restricting only linear correlation can be insufficient in practice and may even reduce usability in certain settings [15]. Alternatively, some approaches employ the Hilbert-Schmidt independence criterion (HSIC) [60] to capture nonlinear dependence. However, HSIC relies on kernel computations and is sensitive to kernel hyperparameters such as the bandwidth σ ; inappropriate choices can bias dependence estimation and reduce robustness across scenarios.

Nonlinear correlation constraint. We observe that shallow neural network features retain significant similarity to the input and are derived through nonlinear operations. To address this, we use distance correlation to measure the non-linear relationship between x and z , incorporating it as a loss constraint. This helps remove redundant input information from the output and effectively increase $H(x | z)$. The calculation of distance correlation is as follows:

$$\mathcal{L}_{\text{d Cor}} = \text{d Cor}(x, z) = \frac{\text{d Cov}(x, z)}{\sqrt{\text{d Var}(x) \cdot \text{d Var}(z)}} \quad (13)$$

Where $\text{d Cov}(x, z)$ represents the distance covariance between x and z , and the denominator is the product of the distance variances for the two samples.

To compute the numerator, we first construct a distance matrix A , where each element is the Euclidean distance between samples, i.e., $A_{ij} = \|x_i - x_j\|_2$. To remove the effect of sample shifts, the distance matrix is double-centred as $A_{ij}^c = A_{ij} - \bar{A}_{i.} - \bar{A}_{.j} + \bar{A}_{..}$, where $\bar{A}_{i.}$ and $\bar{A}_{.j}$ denote the row and column means, respectively, and $\bar{A}_{..}$ denotes the overall mean. Similarly, we construct $B_{ij} = \|z_i - z_j\|_2$ and obtain its double-centred version B_{ij}^c . With the centred distance matrices, the distance covariance is computed as $\text{dCov}(x, z) = \frac{1}{n^2} \sum_{i,j} A_{ij}^c B_{ij}^c$, where n is the batch size.

The distance variance is defined as $\text{d Var}(x) = \frac{1}{n^2} \sum_{i,j} A_{ij}^{c^2}$, which normalizes the distance covariance. The value range of distance correlation is $[0, 1]$, where 1 indicates perfect correlation. Since x and z exhibit high distance similarity and are derived through nonlinear operations, distance correlation is well-suited as a constraint term. Moreover, as a distribution-free measure, it does not rely on specific data distributions, making it effective for non-Gaussian distributions, nonlinear relationships, and complex data types such as images and text. For non-stationary and multimodal data, distance correlation provides stable dependency measurements, allowing the constraint to flexibly adapt to different models and data.

Linear correlation constraint. In addition to constraining the correlation between x and z , it is necessary to design loss constraints specifically for the z . The primary issue lies in the z containing repetitive or similar information, which is redundant for decision-making and provides additional input for Gen-based MIAs, strengthening local constraints in the input space and improving reconstruction accuracy.

To address this, we propose using the Pearson correlation coefficient as a loss constraint to reduce the linear correlation within z , thereby suppressing redundant information; in conjunction with sparsity regularisation, this also reduces $\delta(z)$. The calculation is as follows:

$$\begin{aligned} \mathcal{L}_{\text{Pearson}} &= \frac{1}{C^2} \sum_{c_1=1}^C \sum_{c_2=1}^C (\text{pCor}_{c_1, c_2})^2 \\ &= \frac{1}{C^2} \sum_{c_1=1}^C \sum_{c_2=1}^C \left(\frac{\text{Cov}_{c_1, c_2}}{\sigma_{c_1} \cdot \sigma_{c_2}} \right)^2 \end{aligned} \quad (14)$$

Where Cov_{c_1, c_2} represents the covariance matrix between channels c_1 and c_2 , calculated as $\text{Cov}_{c_1, c_2} = \frac{1}{H \cdot W} \sum_{i=1}^{H \cdot W} (z_{c_1, i} - \mu_{c_1}) \cdot (z_{c_2, i} - \mu_{c_2})$, where $z_{c_1, i}$ is the i -th element of channel c_1 and μ_{c_1} is its mean. H and W denote the height and width of the feature space. σ_{c_1} and σ_{c_2} are the standard deviations, calculated as $\sigma_{c_1} = \sqrt{\frac{1}{H \cdot W} \sum_{i=1}^{H \cdot W} (z_{c_1, i} - \mu_{c_1})^2}$.

To compute the scalar $\mathcal{L}_{\text{Pearson}}$, the squared correlations of all channel pairs are averaged, keeping the value range within $[0, 1]$. This avoids negative values, which could lead to suboptimal updates during gradient descent. Negative values might drive correlations toward complete negative correlation, introducing unnecessary redundancy or unsuitable learning patterns and hindering convergence. Therefore, the goal is to minimize the coefficient toward zero, as lower values indicate reduced linear correlation. Using the squared correlations and

their average ensures a smoother and more stable gradient descent process.

Additionally, to prevent extreme values in the feature output weights during the optimization of the above constraints, we include the l_1 -norm as a loss term. This term is added specifically to suppress weights, but excessive emphasis could interfere with normal model training. Therefore, the coefficient τ for this loss term is set below $1e-2$ to balance its influence.

Label smoothing. Finally, to ensure model usability, we draw inspiration from [33] and apply LS to refine the target loss, as shown in formula (15). Additionally, KL divergence is used instead of cross-entropy (CE) as the primary loss function for optimization.

$$LS(\hat{y}_c) = 1 - \alpha + \frac{\alpha}{K}, \quad LS(\hat{y}_i) = \frac{\alpha}{K} \quad (15)$$

Where $LS(\hat{y}_c)$ refers to the label smoothing element under the target class, $LS(\hat{y}_i)$ is the i -th element of the non-target class. α is the smoothing factor and K is the total number of categories.

While the z at this layer has not fully transitioned into decision-level semantics, the combination of architectural constraints and the implicit regularization of $I(x; z)$ imposed by the loss function encourages more task-relevant representations. Consequently, positive-factor LS can induce slight intra-class compactness, modestly reducing the feature entropy and hindering inversion attacks [33], as evidenced in Section VI-C.

Even in cases where LS increases entropy due to insufficient decision-oriented expression, its impact on attack vulnerability remains limited. Ultimately, label smoothing is intended to promote generalization, with improved resistance to MIAs appearing only as a byproduct when sufficient constraints are imposed on the feature representations.

In summary, the proposed loss function first applies non-linear correlation constraints to increase $H(x|z)$ and suppress $H(z)$. Next, it applies linear correlation and extreme value constraints on z to further enhance $H(x|z)$ and suppress $\delta(z)$. Finally, LS and KL divergence maintain usability while slightly suppressing $H(z)$. The final formulation of this optimization is as follows, with specific parameter settings.

$$\begin{aligned} \min_{\theta} \mathcal{L}_{\theta}(x, z, \hat{y}) = & \lambda_1 \cdot KL(f(x), LS(\hat{y})) + \lambda_2 \cdot \mathcal{L}_{\text{d Cor}} \\ & + \lambda_3 \cdot \mathcal{L}_{\text{Pearson}} + \tau \cdot \|z\|_1. \end{aligned} \quad (16)$$

Specific parameter settings are referenced in Section VI-A. SiftFunnel and MINE are available at <https://github.com/SiftFunnel/SiftFunnel>.

VI. EXPERIMENTS

This section evaluates SiftFunnel under diverse datasets, model architectures, and attack settings. We compare SiftFunnel with representative perturbation-based, IB-based, and NND & IB-based defences, and we consider both MLE-based and Gen-based model inversion attacks, including gray-box settings when applicable. We report metrics that cover usability, privacy, and deployability, including test accuracy, reconstruction quality, estimated mutual information via MINE, $\delta(z)$, edge parameter count, and inference latency on CPU and

TABLE I
DATA ALLOCATION OF THE TARGET MODEL AND ATTACK MODEL.

Target model		Attack Model
Task	Data	Auxiliary Data
CIFAR-10 (10 classes)	Train: 66.6%, Test: 16.7%	16.7% (10 classes of CIFAR-10)
FaceScrub (530 classes)	Train: 80%, Test: 20%	CelebA (non-individual overlapping)
ChestX-ray (4 classes)	Train: 66.6%, Test: 16.7%	16.7% (4 classes of ChestX-ray)

GPU. The trends of estimated mutual information and $\delta(z)$ provide empirical support for the proposed difficulty criterion. We further provide ablation studies to isolate the impact of the funnel design, attention blocks, and each loss component.

A. Experimental Setup

1) *Datasets.* We used three types of image recognition datasets, with processing details outlined as follows. The detailed data allocation is shown in Table I.

- **CIFAR-10** [51]. The CIFAR-10 dataset consists of 60,000 RGB images categorized into ten classes with an original resolution of 32×32 . To make the results clearer and more intuitive, we used the public 64×64 release¹. Moreover, the CAI-super-resolution procedure is not simple pixel duplication or naive interpolation.
- **FaceScrub** [61]. FaceScrub is a URL dataset with 100,000 images of 530 actors, which contains 265 male actors and 265 female actors. However, since not every URL was available during the writing period, we downloaded a total of 43,149 images for 530 individuals and resized the images to 64×64 .
- **CelebA** [62]. CelebA is a dataset with 202,599 images of 10,177 celebrities. We used the same crop as [24] to remove the background of images in this dataset other than faces to reduce the impact on the experiment. To eliminate individual overlap, we removed a total of 6,878 images of 296 individuals and similarly resized the images to 64×64 .
- **ChestX-ray** [63]. This dataset is a curated collection of COVID-19 chest X-ray images compiled from 15 publicly available datasets. It contains 1,281 COVID-19 X-rays, 3,270 normal X-rays, 1,656 viral pneumonia X-rays, and 3,001 bacterial pneumonia X-rays with a resolution of 128×128 .

2) *Attack Method.* The selection of attack methods has been discussed in Section III-B. Specifically, for MLE-based MIA, this paper has adjusted to the hyper-parameter values that yield the best results. For Gen-based MIA, the attack model in this paper is designed as the inverse architecture of the edge model. Moreover, under the gray-box scenario assumption, the attack model can enhance the expressive power of the reverse mapping as the edge model architecture changes. The attack model is trained using the Adam optimizer with a learning rate of $2e-4$, a β_1 of 0.5, and a ReduceLROnPlateau scheduler with a factor of 0.5 and patience of 20. The attack model training is based on strong assumptions, as shown in Table 1,

¹<https://www.kaggle.com/datasets/joapauloschuler/cifar10-64x64-resized-via-cai-super-resolution>

TABLE II
QUANTITATIVE EVALUATION OF CNN ON CIFAR-10 WITH DIFFERENT DEFENCES AGAINST TWO MIAS.

Method Class	Method	Test ACC↑	MLE-based Attack			Gen-based Attack			MI↓	$\delta(z)\downarrow$	$ \theta_{\text{edge}} \downarrow$
			MSE↑	PSNR↓	SSIM↓	MSE↑	PSNR↓	SSIM↓			
Unprotected		88.28%	0.0045	28.3421	0.9883	0.0017	32.5498	0.9987	0.0566	32,549	299,520
Perturbation-based	Adding noise [13] ($\sigma = 0.8$)	74.85% (-13.43%)	0.0291	20.1511	0.9135	0.0172	22.4249	0.9299	0.0502	65,535	299,520
	Dropping [13] ($r = 0.8$)	21.18% (-67.10%)	0.0127	23.7340	0.9609	0.0135	23.4835	0.9481	0.0501	6,509	
	DPSGD [64] (20, 1e-5)	61.83% (-26.45%)	0.0218	21.3868	0.9479	0.0016	32.4800	0.994	0.0623	39,829	
	NLS [33] ($\alpha = -0.05$)	85.53% (-2.75%)	0.0002	42.2684	0.9996	0.0015	33.0003	0.9943	0.0573	47,183	
IB-based	MID [34] (1e-2)	81.53% (-6.75%)	1.74e-5	50.4675	0.9999	0.0013	34.5184	0.9967	0.0584	45,198	299,520
	BiDO [15] (2; 20)	84.12% (-4.16%)	0.6699	6.5111	0.0138	0.035	19.2965	0.8751	0.0429	3,345	
	VIB [14]	88.21% (-0.07%)	0.5552	7.3264	0.0358	0.0042	28.5054	0.9842	0.0551	18,069	
	InfoSCISSORS [18]	83.46% (-4.83%)	0.5715	7.2007	0.1345	0.0062	26.8799	0.9948	0.0549	43,364	
NND & IB-based	AE-based [11]	90.56% (+2.28%)	0.6345	6.7472	0.0051	0.0542 0.01926	17.4852 21.9259	0.8151 0.9295	0.0358	2,617	2,365,056
	SiftFunnel	85.49% (-2.79%)	0.6792	6.4510	0.0051	0.1166 0.0639	14.1112 16.7314	0.5419 0.7647	0.0167	1,006	14,911

Note: $\uparrow$ indicates higher is better, $\downarrow$ indicates lower is better. Bolded values are the best. Blue highlights show unprotected baseline performance, red indicates increased edge device load, yellow indicates reduced load, and gray shows gray-box defence performance.

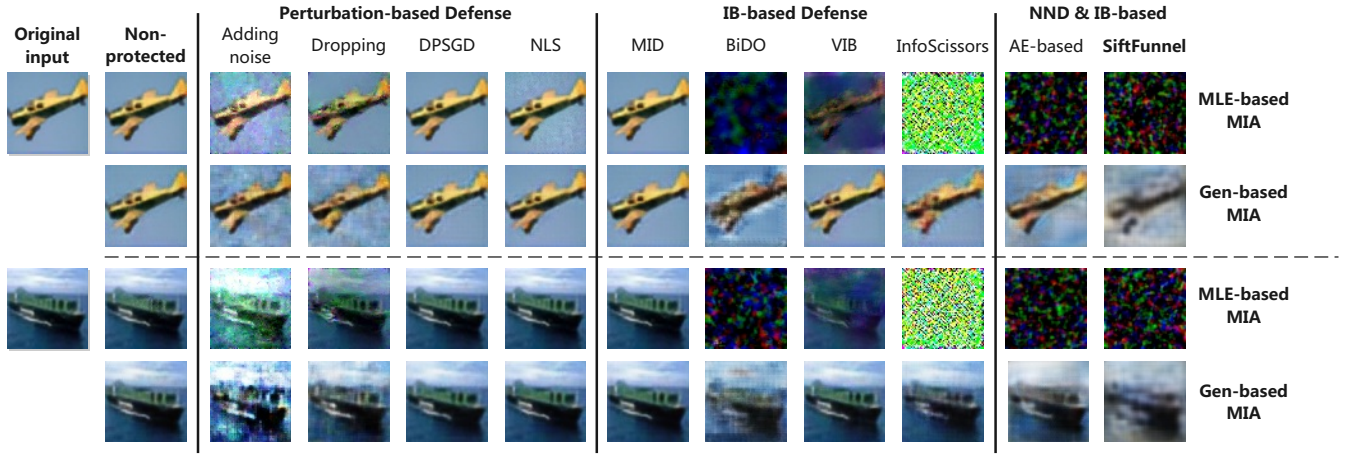


Fig. 4. A visual evaluation of the attack effectiveness of two MIAs on CNN edge models protected by various defence techniques, with NND & IB-based results specifically reflecting the gray-box scenario.

and only the models with the minimum test MSE are saved to highlight the defence's effectiveness.

3) *Target Edge Model and Implementation Details.* We employ the same comprehensive model architecture as detailed in [24], comprising four CNN blocks succeeded by two fully connected layers. This model is trained using the Adam optimizer with a learning rate of $2e-4$, a β_1 parameter of 0.5, and a ReduceLROnPlateau scheduler with a reduction factor of 0.5 and patience of 25. Each CNN block is composed of a convolutional layer, a batch normalization layer, a max-pooling layer, and a ReLU activation function. The target edge model encompasses the initial two CNN blocks. For the training of the edge model protected by SiftFunnel, we have specified the parameters λ_1 to 3.5, α to 0.35, λ_2 to 0.8, and λ_3 to 0.6 in Equation (15). In addition, to argue the impact of the model architecture for the defence, we use different CNNs as follows: (1) VGG16 adapted from [65]; (2) ResNet-18 adapted from [38], the learning rates are $3e-4$. The division positions of ResNet and VGG16 are not fixed. For detailed accuracy and how to divide them, see Section VI-C for analysis. All experiments were performed on two RTX 4090 GPUs and an

Intel Core i9-14900KF $\times$ 32 CPU.

4) *Comparison of defence Methods.* We aim to compare three existing categories of defence methods. For Perturbation-based defences, we select **Adding noise & Dropping features** [13], **DPSGD** [64], and negative label smoothing (**NLS**) [33]. For IB-based defences, we choose **MID** [34], **BiDO** (with HSIC) [15], **VIB** [14], and **InfoSCISSORS** [18]. For NND & IB-based defences, we opt for **AE-based protection** [11]. Our method falls into the NND & IB-based defence category, which also requires analysing the defensive effect in a gray-box scenario. Furthermore, all comparison methods have been selected and set with appropriate hyper-parameters for comparison, with detailed settings found in Section VI-C. Lastly, all methods are trained without adversarial training, focusing solely on the effects of the original defence methods.

B. Evaluation Metrics

We selected 7 evaluation metrics to assess the aspects of usability, privacy, and deployability. For usability assessment, we employ **Test Accuracy (Test ACC)**, which evaluates the final performance of the model protected by defence techniques on the test dataset. Regarding privacy assessment, we

TABLE III

QUANTITATIVE EVALUATION OF RESNET-18 ON CIFAR-10 WITH DIFFERENT DEFENCES AGAINST TWO MIAs. THE TARGET EDGE MODEL IS THE FIRST RESIDUAL BLOCK OF RESNET-18.

Method Class	Method	Test ACC↑	MLE-based Attack			Gen-based Attack			MI↓	$\delta(z)\downarrow$	$ \theta_{\text{edge}} \downarrow$
			MSE↑	PSNR↓	SSIM↓	MSE↑	PSNR↓	SSIM↓			
Unprotected		90.43%	0.0001	71.0996	0.9991	0.0010	34.6599	0.9991	0.9719	182,134	149,824
Perturbation-based	Adding noise [13] ($\sigma = 2$)	81.39% (-9.04%)	0.0035	29.3354	0.9872	0.0054	27.4167	0.9794	0.9619	262,144	149,824
	Dropping [13] ($r = 0.8$)	14.09% (-76.34%)	0.0069	26.3641	0.9754	0.0077	25.9334	0.9711	0.9514	35,574	
	DPSGD [64] (20, 1e-5)	61.85% (-28.58%)	0.0115	23.9453	0.9628	0.0060	26.9751	0.9802	0.9515	178,956	
	NLS [33] ($\alpha = -0.05$)	88.13% (-2.30%)	0.0034	29.5001	0.9894	0.0008	36.0306	0.9974	0.9721	176,456	
IB-based	MID [34] (1e-2)	79.48% (-10.95%)	1.42e-7	73.2595	0.9999	0.0010	34.5893	0.9965	0.9729	173,587	149,824
	BiDO [15] (5; 10)	90.08% (-0.35%)	4.38e-7	68.3603	0.9999	0.0020	31.7621	0.9927	0.9647	219,832	
	VIB [14]	89.18% (-1.25%)	0.5869	7.0853	0.0216	0.0020	31.7621	0.9928	0.9402	54,800	
	InfoSCISSORS [18]	88.03% (-2.40%)	3.10e-5	49.8548	0.9999	0.0058	27.1608	0.9799	0.8927	216,919	
NND & IB-based	AE-based [11]	89.19% (-1.24%)	0.6543	6.6189	0.1243	0.0143 0.0094	23.2332 25.0633	0.9488 0.9660	0.9397	74,356	279,136
	SiftFunnel	88.43% (-2.00%)	0.6538	6.6166	0.0059	0.1957 0.2790	11.8562 10.3230	0.0793 0.3001	0.4772	4,028	78,864

Note: $\uparrow$ indicates higher is better, $\downarrow$ indicates lower is better. Bolded values are the best. The meanings of colour-coded regions are consistent with those in Table II.

TABLE IV

QUANTITATIVE EVALUATION OF DIFFERENT DEFENCE METHODS IN PROTECTING CNN AGAINST MIAs ON THE FACE SCRUB DATASET.

Method Class	Method	Test ACC $\uparrow$	MLE-based Attack			Gen-based Attack			MI $\downarrow$	$\delta(z)\downarrow$	$ \theta_{\text{edge}} \downarrow$
			MSE $\uparrow$	PSNR $\downarrow$	SSIM $\downarrow$	MSE $\uparrow$	PSNR $\downarrow$	SSIM $\downarrow$			
Unprotected		88.31%	0.0005	37.6115	0.9972	0.0006	32.4108	0.9908	0.0514	45,271	299,520
Perturbation-based	NLS [33] ($\alpha = -0.2$)	88.03% (-0.28%)	0.0004	38.8904	0.9980	0.0017	32.5480	0.9911	0.0515	44,783	299,520
IB-based	MID [34] (1e-2)	79.45% (-10.98%)	0.0017	32.3435	0.9897	0.0018	32.1285	0.9901	0.0507	39,277	299,520
	BiDO [15] (2; 20)	88.02% (-0.29%)	0.0004	39.0838	0.9978	0.0018	32.2048	0.9904	0.0493	44,915	
	VIB [14]	88.05% (-0.26%)	0.7354	6.1060	0.0032	0.0033	29.5697	0.9823	0.0493	15,649	
	InfoSCISSORS [18]	81.61% (-6.70%)	0.7449	6.0504	0.0026	0.0060	26.9803	0.9675	0.0408	57,050	
NND & IB-based	AE-based [11]	83.45 (-4.86%)	0.6474	6.6594	0.0090	0.0609 0.0220	16.9260 21.3444	0.6656 0.8795	0.0238	585	2,365,056
	SiftFunnel ($\lambda_1 = 5.0, \lambda_2 = 0.1, \lambda_3 = 0.1$)	85.39% (-2.92%)	0.7264	6.1593	0.0060	0.0559 0.0296	17.3018 20.0576	0.6854 0.8323	0.0104	637	14,911

Note: Changes in data types alter the inference task, requiring adjustments to the parameters of the defence methods. The modified parameters are listed in the table's second column.

initially utilize **Mean Squared Error (MSE)**, **Peak Signal-to-Noise Ratio (PSNR)**, and **Structural Similarity Index (SSIM)** to evaluate the effectiveness of MIA on reconstructing the input data of the protected model [8]. Here, a higher MSE and lower PSNR and SSIM (ranging from 0 to 1) indicate better defence effectiveness. In addition, we employ MINE [19], as mentioned in Section IV-C, to report two privacy-indicative metrics, **Mutual Information (MI)** and the **Effective Information Mean $\delta(z)$** . We report MI as a MINE-based normalized score on the unit interval from 0 to 1. Concretely, our ConvMINE critic takes the channel-wise concatenation of x and z as input, outputs a score map through convolutional layers, applies a Sigmoid activation to constrain the score to $[0, 1]$, and then averages the score over channels and spatial locations to obtain a per-sample scalar, which is finally averaged over the test set. We define $\delta(z)$ as the test-set average number of non-zero elements in the edge output. When comparing different defence methods under the same estimator and protocol, lower estimated MI and lower $\delta(z)$ indicate that the transmitted representation carries less input-related information and is harder to invert. The training procedure and the architecture are provided at <https://github.com/SiftFunnel/SiftFunnel>. For deployability, we consider the **Edge model's parameter count** $|\theta_{\text{edge}}|$ and Inference Latency. Model parameters directly affect memory

usage, and latency is measured on both CPU and GPU.

C. Experimental Results

The results are analysed from 4 perspectives: the impact of defence methods on $I(x; z)$ and $\delta(z)$, the influence of datasets and model architectures, and ablation studies.

1) *The effect of various defences on the MI and $\delta(z)$.* As shown in Table II, defence effectiveness is reflected by MI and $\delta(z)$, aligning with the criterion D_{mia} . Specifically, unprotected edge models are highly vulnerable to MIAs, with MI at 0.0566, MSE below 0.01, and SSIM exceeding 0.98. Perturbation-based defences reduce MI by at most 11% but cause over a 10% drop in Test ACC, with NLS even increasing MI in shallow networks, weakening resistance to attacks. From the theoretical analysis, it can be seen that DPSGD does not alter the complexity of the mapping from x to z , nor does it directly affect the intermediate feature z itself. As a result, it reduces model accuracy without effectively mitigating MIA risks in the CI setting. IB-based defences slightly reduce usability but effectively counter MLE-based MIAs, except for MID, which shows weaker resistance. Among them, BiDO significantly impacts MI and $\delta(z)$, providing effective defence, but its parameter selection and kernel width configuration (as shown in the table) limits generalizability, as confirmed in subsequent experiments. This experiment follows the InfoS-

TABLE V

QUANTITATIVE EVALUATION OF VGG-16 DEFENCE EFFECTIVENESS ACROSS DIFFERENT DATASETS, USING GEN-BASED MIA AS THE ATTACK METHOD.

Datasets	Method	ACC $\uparrow$	MSE $\uparrow$	PSNR $\downarrow$	SSIM $\downarrow$	MI $\downarrow$	$\delta(z)\downarrow$	$ \theta_{\text{edge}} \downarrow$
CIFAR-10	Unprotected	91.07%	0.0057	27.2557	0.9787	0.0515	31,241	556,608
	AE-based [11]	90.21% (-0.86%)	0.0179	22.2649	0.9318	0.0442	15,436	3,064,896
	SiftFunnel	88.21% (-2.86%)	0.0923	15.1347	0.6151	0.0174	1,852	306,630
FaceScrub	Unprotected	89.98%	0.0080	25.7434	0.9542	0.0386	21,513	556,608
	AE-based [11]	87.70% (-2.28%)	0.0216	21.4474	0.8840	0.0309	14,119	3,064,896
	SiftFunnel	84.70% (-5.28%)	0.0917	15.1720	0.5211	0.0148	255	300,582
ChestX-ray	Unprotected	89.57%	0.0010	30.1644	0.9770	0.0262	133,295	555,456
	AE-based [11]	87.83% (-1.74%)	0.0021	26.8429	0.9475	0.0219	62,917	3,063,744
	SiftFunnel	87.61% (-1.96%)	0.0311	15.0999	0.1608	0.0015	1,023	299,430

Note: The partition is applied at the fourth ReLU layer of VGG-16. The Gen-based attack results were derived from a gray-box setting. For CIFAR-10, the hyperparameters of SiftFunnel follow the setting in Section VI-A, item 3). For FaceScrub, the allocation is set to $(\lambda_1 = 4.5, \lambda_2 = 0.4, \lambda_3 = 0.1)$, and for ChestX-ray, it is $(\lambda_1 = 3.5, \lambda_2 = 1.2, \lambda_3 = 0.8)$.

TABLE VI

QUANTITATIVE EVALUATION OF MIA RESISTANCE ON CIFAR-10 ACROSS DIFFERENT ARCHITECTURES.

Architecture	Test ACC $\uparrow$	MLE-based Attack			Gen-based Attack			MI $\downarrow$	$\delta(z)\downarrow$	$ \theta_{\text{edge}} \downarrow$
		MSE $\uparrow$	PSNR $\downarrow$	SSIM $\downarrow$	MSE $\uparrow$	PSNR $\downarrow$	SSIM $\downarrow$			
CNN	85.49% (-2.79%)	0.6792	6.4510	0.0051	0.1166	14.1112	0.5419	0.0167	1,006	14,911
ResNet-18	88.43% (-2.00%)	0.6538	6.6166	0.0059	0.0639	16.7314	0.7647	0.4772	4,028	78,864
VGG16	88.89% (-2.18%)	0.7801	5.8450	0.0005	0.1957	11.8562	0.0793	0.0128	3,587	52,086

Note: The configurations for the CNN and ResNet-18 edge models are provided in Tables II and III. For the VGG-16 architecture, the split is implemented at the first ReLU layer immediately after the max-pooling operation.

CISSORS setup, ensuring CLUB receives a complete dataset and sufficient training for convergence. While the CLUB effectively defends against MLE-based MIAs, it struggles against Gen-based MIAs and incurs a 4.83% accuracy loss. AE-based defences improve model usability, due to enhanced model capacity, while MI reductions confirm reasonable defence effectiveness. However, gray-box scenarios weaken AE-based defences, and their added complexity increases storage costs by $8\times$, as seen in $|\theta_{\text{edge}}|$. SiftFunnel achieves the best defence performance against both MIAs while keeping accuracy loss below 3%. It achieves MSE of 0.6792 (MLE-based MIA) and above 0.1 (Gen-based MIA). In gray-box scenarios, it maintains MSE of 0.06 and SSIM of 0.7647, ensuring robust resistance. Furthermore, MI and $\delta(z)$ confirm its optimal impact on the key factors of D_{mia} . Additionally, $|\theta_{\text{edge}}|$ reveals a nearly $20\times$ reduction in edge model load, highlighting SiftFunnel's efficiency.

2) *The effect of model architecture.* Modifying the model architecture does not alter the parameter settings of the proposed method, which remain consistent with Section VI-A. We adapt the model to ResNet-18 with skip connections, using the first residual block as the edge model. This setup poses challenges for defence techniques, as shallow networks retain simple structures and do not modify spatial dimensions. Table III shows that unprotected models exhibit minimal MI impact, resulting in weak resistance to MIAs, with MLE-based MIA achieving an MSE of 10^{-4} and Gen-based MIA 10^{-3} . Under these conditions, perturbation-based defences fail to balance usability and privacy, leading to poor defence performance. Among IB-based defences, only VIB demonstrates resistance to MLE-based MIA by optimizing the edge model through $I(x; z)$ estimation as a loss function, but its effectiveness

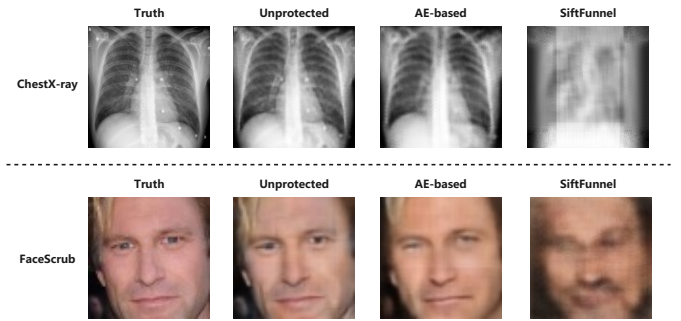


Fig. 5. Visual evaluation of privacy and usability against Gen-based MIA in the VGG16 edge model on the ChestX-ray $(\lambda_1 = 3.5, \lambda_2 = 1.2, \lambda_3 = 0.8)$ and FaceScrub $(\lambda_1 = 4.5, \lambda_2 = 0.4, \lambda_3 = 0.1)$.

TABLE VII

QUANTITATIVE EVALUATION OF SIFTFUNNEL'S DEFENCE PERFORMANCE ACROSS DIFFERENT DATASETS AND MODEL ARCHITECTURES.

Dataset	Model	Test ACC $\uparrow$	MLE-based Attack			Gen-based Attack			MI $\downarrow$	$\delta(z)\downarrow$	$ \theta_{\text{edge}} \downarrow$
			MSE $\uparrow$	PSNR $\downarrow$	SSIM $\downarrow$	MSE $\uparrow$	PSNR $\downarrow$	SSIM $\downarrow$			
FaceScrub	CNN	85.39% (-2.92%)	0.7264	6.1593	0.0060	0.0296	20.0576	0.8323	0.0104	637	14,911
	ResNet-18	88.58% (-2.31%)	0.6539	6.6158	0.0124	0.0398	18.7835	0.7894	0.0513	1,872	378,204
	VGG-16	84.70% (-5.28%)	0.7205	6.1950	0.0079	0.0917	15.1720	0.5211	0.0148	255	300,582
ChestX-ray	CNN	88.70% (-0.54%)	0.2310	6.3641	0.0186	0.0335	14.7713	0.1369	0.0061	1,022	3,330
	ResNet-18	86.20% (-2.93%)	0.3329	4.7765	0.0178	0.3076	5.1199	0.0031	0.0056	231	377,792
	VGG-16	87.61% (-1.96%)	0.2333	6.3215	0.0055	0.0311	15.0999	0.1608	0.0015	1,023	299,430

Note: Gen-based attack results were obtained under a gray-box setting. For ResNet-18, we partition after Block-1 on ChestX-ray with $\lambda = (\lambda_1, \lambda_2, \lambda_3) = (3.5, 0.8, 0.6)$, and after Block-2 on FaceScrub with $\lambda = (2.2, 1.65, 0.12)$.

remains limited, particularly against Gen-based MIA. For BiDO, despite applying kernel bandwidth estimation methods from [15] and extensive experimental tuning, it fails to achieve a configuration that effectively balances usability and privacy, resulting in suboptimal defence performance. InfoSCISSORS impacts MI under this configuration, but the increase in $\delta(z)$ undermines its effectiveness against Gen-based MIAs, leading to only marginal improvements. AE-based defences show some resistance to attacks on low-dimensional features in earlier stages, but their effectiveness diminishes under the ResNet configuration. While they reduce $\delta(z)$ by 60%, they have minimal impact on MI, achieving only a 0.04% reduction. Consequently, AE-based defences struggle to counter both black-box and gray-box Gen-based MIAs effectively.

In contrast, SiftFunnel reduces MI by 50% and lowers $\delta(z)$ to the millesimal scale, offering the strongest MIA resistance while maintaining usability. It achieves a minimum SSIM of 0.0793, highlighting its privacy-utility balance. Table VI compares performance across architectures.

TABLE VIII

SIFTFUNNEL ABLATION STUDY ON CIFAR-10. THE TARGET EDGE MODEL IS THE FIRST RESIDUAL BLOCK OF RESNET-18, WITH GEN-BASED MIA AS THE ATTACK METHOD. THE RED AREA INDICATES THE STRONG ATTACK HYPOTHESIS FROM SECTION V-B.

Method	Test ACC $\uparrow$	MSE $\uparrow$	PSNR $\downarrow$	SSIM $\downarrow$	MI $\downarrow$	$\delta(z)\downarrow$
Unprotected	90.43%	0.0010	34.6599	0.9991	0.9719	182,134
Without Funnel (Average training round takes 1 minute, batch size=64)	90.33% (-0.10%)	0.0594	17.0709	0.7776	0.8945	24,322
Without Attention Blocks	88.45% (-1.98%)	0.0525	17.5709	0.8178	0.8763	3,904
Without KL and LS	88.59% (-1.84%)	0.0850	15.4813	0.6905	0.7935	3,994
Without $\mathcal{L}_{dCor}$	88.77% (-1.66%)	0.0268	20.4929	0.9025	0.9505	4,013
Without $\mathcal{L}_{Pearson}$	88.83% (-1.60%)	0.0492	17.8576	0.8294	0.8618	4,037
Without l_1	88.48% (-1.95%)	0.0789	15.8057	0.7272	0.8158	4,030
Attack SiftFunnel in Cloud	88.43% (-2.00%)	0.0521	17.6159	0.8205	0.8905	176,380
SiftFunnel		0.2790	10.3230	0.3001	0.4772	4,028

3) *The effect of data type.* Changes in data types lead to shifts in task requirements. For instance, in this study, the facial recognition task involves 530 classes, while ChestX-ray classification involves 4 classes, necessitating parameter adjustments as shown in Table IV. As analysed in the previous section on model architectures, to achieve better defence performance, we adjusted the edge model divisions for ResNet-18 and VGG. Specifically, the division points were set after the second residual block for ResNet and after the first ReLU layer following the second max-pooling layer for VGG. The results, summarized in Table IV and Figure 5, show that the defence performance remains unaffected by changes in data types. SiftFunnel continues to be the best defence method, offering an optimal balance between usability, privacy, and deployability.

As shown in Table V and Figure 5, the SiftFunnel-protected VGG16 model demonstrates superior defence against Gen-based MIA. This is evident not only in the consistently low PSNR values (around 15) on facial and X-ray data, which indicate a significant degradation in the visual quality of reconstructed images, but also in its ability to effectively reduce both the $\delta(z)$ and $|\theta_{edge}|$ across all datasets. This consistent reduction confirms SiftFunnel's robust deployability, ensuring strong privacy protection regardless of the specific dataset.

To provide a comprehensive quantitative view of our evaluation, we additionally include Table VII, which reports SiftFunnel's results across different data types and architectures.

4) *Ablation Study.* To assess the effectiveness of SiftFunnel's model structure, loss function design, and defence capability, we conducted ablation experiments using Gen-based MIA. As shown in Table VIII, we separately removed the Information Funnel and Attention Blocks from the model structure. The results indicate that removing the Funnel had minimal impact on accuracy, reducing it by only 0.1%, whereas removing the attention mechanism significantly increased MI, weakening privacy protection. For the loss function, we examined the effect of removing KL divergence, linear and nonlinear constraints, and the l_1 -norm. The results show that KL divergence and l_1 -norm provided moderate defence improvements, while nonlinear constraints were critical—removing them increased

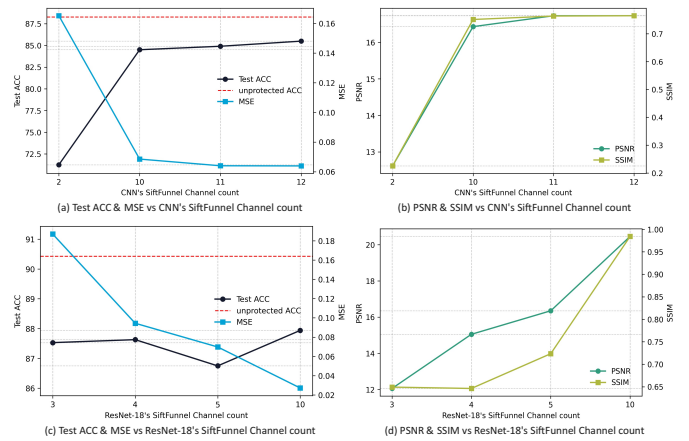


Fig. 6. Ablation study of SiftFunnel's Channel count on Cifar-10. CNN is under the spatial-size-reduced setting, while ResNet-18 is under the spatial-size-unchanged setting.

MSE to 0.0268. Additionally, we validated the hypothesis from Section V-B, demonstrating that even when adversaries reconstructed inputs using cloud-compensated features, SiftFunnel still outperformed existing defences in protecting x .

Figure 6 presents the ablation results for varying the number of output channels while keeping all other training hyperparameters constant. As shown in Figure 6 (a,b), when the spatial size is reduced, a very low channel count of 2 offers the strongest privacy protection but is impractical due to a significant accuracy drop of 17.02%. The corresponding change from 10 to 12 channels shows a gradual improvement in accuracy with a slight trade-off in privacy. Figures 6 (c,d) demonstrate that a low channel count is feasible when the spatial size remains unchanged. It is important to note that since we did not alter the hyperparameters in Equation (15), a high channel count results in the gradient optimization from the correlation constraint outweighing the gradient for optimizing utility. Therefore, the hyperparameter allocation in practical training needs to be adjusted accordingly. All other observed effects align with our intuitive predictions.

We further evaluate the inference latency of edge models on both CPU and GPU, and summarize the results in Table IX.

TABLE IX
INFERENCE LATENCY (MS) OF EDGE MODELS ON CPU AND GPU.

Architecture	Device	SiftFunnel	Unprotected	AE-based
CNN	CPU	3.67	6.15	13.33
	GPU	0.49	0.28	0.78
ResNet-18	CPU	13.96	14.32	20.68
	GPU	0.92	0.51	0.89
VGG-16	CPU	7.46	8.88	15.68
	GPU	0.78	0.33	0.74

Note: Latency is reported in milliseconds (ms). Mean latency is computed over 200 iterations after 50 warmup runs. CPU uses one thread. GPU timing uses CUDA events with synchronization. Bold text indicates the lowest latency in each row.

Latency is computed by timing 200 repeated executions of the edge forward pass after 50 warmup runs and averaging the recorded times. On CPU, we report single-sample latency with one thread. On GPU, we run 10 samples together per forward pass to obtain stable timing, since single-sample GPU timing is typically sensitive to kernel launch overhead and runtime variance, and GPU parallelism supports batch execution. Inputs are created directly on the target device so that data transfer is excluded.

Table IX shows that SiftFunnel consistently maintains competitive latency across architectures while providing substantially stronger privacy protection. On CPU, SiftFunnel is the fastest option for all architectures, reducing latency from 6.15 ms to 3.67 ms on CNN, from 14.32 ms to 13.96 ms on ResNet-18, and from 8.88 ms to 7.46 ms on VGG-16. On GPU, the unprotected model is the fastest, while SiftFunnel remains below 1 ms across all architectures and does not incur the large memory storage burden introduced by AE-based protection. This CPU-GPU difference is expected: CPU latency tends to be more sensitive to parameter footprint and memory access, which SiftFunnel reduces, whereas GPU latency in the sub-millisecond regime can be more influenced by fixed kernel-launch costs and operator fragmentation. Since the funnel structure and attention blocks introduce additional operators, this can lead to a slightly higher GPU latency than the unprotected baseline. Overall, these results indicate that SiftFunnel improves privacy with a lightweight design, and its reduced edge-side complexity also benefits CPU latency in practical settings.

VII. DISCUSSION

The experimental results presented here are not selectively chosen. As detailed in Table VIII, the defence's performance can be maintained above an empirically established lower bound, allowing for practical training adjustments to suit specific scenarios. In this section, we further discuss adaptive attacks and the broader applicability of our criterion.

A. Adaptive attacks

Under the assumptions of a gray-box attack and the attacker's model conditions, our proposed SiftFunnel serves as an effective defence against adaptive attacks. Our results

show that SiftFunnel exhibits a stronger resistance to these attacks compared to the AE-based defence. Furthermore, we conducted a validation of the adaptive attack assumption by testing the attack after integrating a supplementary module on the cloud side, as detailed in Section V-B and Table VIII. This validation confirmed that even with this enhancement, our approach maintains a superior defence capability compared to existing methods.

Building upon the theoretical analysis in Section IV and the guidance from D_{mia} , we can also infer that an adversary could theoretically improve their attack by actively increasing $H(z)$. Addressing this and developing a targeted defence will be a focus of our future work.

B. Theory Applicability

The theoretical analysis in this paper explores the essence of MIAs in CI scenarios. Based on this, we derived the D_{mia} and validated it through both theoretical and experimental analysis. Therefore, this criterion is applicable for evaluating the difficulty of MIAs in other split-processing scenarios, such as split learning [66], semantic communications [14], and real-world facial feature matching. However, it may not be suitable for evaluating MIA on models deployed centrally. In settings where the deployed system introduces randomness in the output (e.g., sampling-based generation or randomized mechanisms), the mapping from X to Y is stochastic and thus $H(Y|X) > 0$. Developing a criterion with a wider range of applicability will be a focus of our future work.

VIII. CONCLUSION

While CI enhances efficiency, it also exposes transmitted features to MIAs, posing severe privacy risks. Despite extensive research, a fundamental understanding of the factors determining MIA effectiveness remains lacking, and existing defences struggle to balance usability, privacy, and deployability, facing theoretical, methodological, and practical constraints. To address these challenges, we proposed the first criterion for evaluating MIA difficulty in CI, identifying MI, entropy, and effective information volume as key influencing factors. Based on this, we proposed SiftFunnel, a privacy-preserving framework designed to limit redundant information while maintaining usability and deployability. By integrating linear and nonlinear correlation constraints, label smoothing, and a funnel-shaped edge model with attention mechanisms, SiftFunnel achieved robust privacy protection with an average accuracy loss of around 3%, and revealed a nearly 20× reduction in edge burdens. Experimental results demonstrated that SiftFunnel offers superior privacy protection across various evaluation metrics and achieves an optimal balance among usability, privacy, and practicality.

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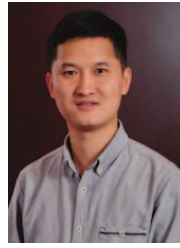
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